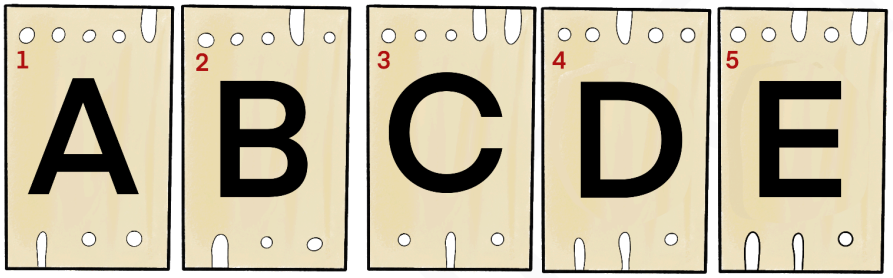
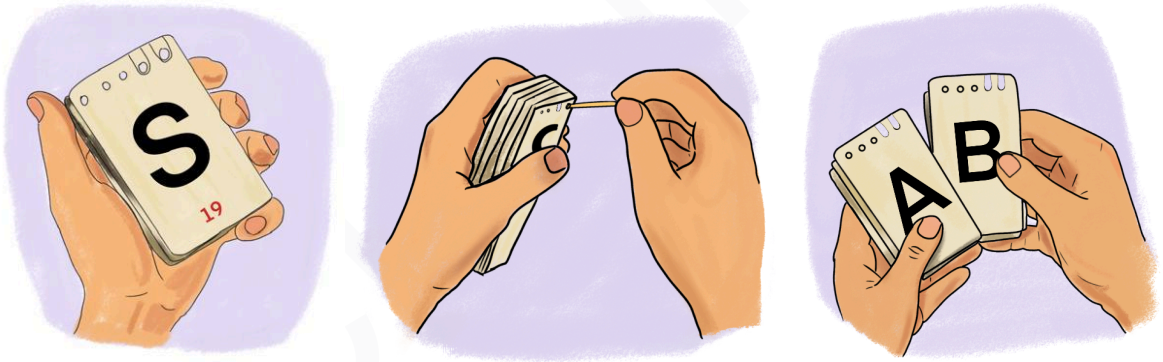
Binary Punch Cards

Grade 7: Peak Beyond A Point

Introduction

We know that numbers can be represented in powers of 10. Though this is not the only way to represent the number. In this activity, we will understand binary representation and powers of 2 using punched cards. Consider a set of cards with the alphabet written on them, and we shuffle them. Using a stick and holes on the top of the cards, we can magically arrange the cards alphabetically in a few seconds. With this activity, we will discover the patterns in punching & the principle behind the trick.





Activity	Time	Description
Launch	5 min	Teacher demonstrate the activity Supporting Links: Print file: https://tinyurl.com/punchedcardstemplate Activity Video: https://tinyurl.com/punchedcardsvideo Reference: https://tinyurl.com/punchedcardsmodule
Trial by Students	15 min	Student's Tryout the activity. Student Worksheet: https://tinyurl.com/punchedcardsworksheet

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		As the teacher progresses through the activities, the students progress along with the worksheet. Students can make their own punch cards using the template after class hours.
Discussions and Explorations	10 min	Attempting the worksheets and the discussion based on the activity

CT Components

Algorithmic Thinking: We used a step-by-step approach to convert a decimal number to binary.

Decomposition: Breaking the conversion of the number to binary into one bit at a time.

Abstraction: Using different types of holes to represent the binary number system.

Logic: Every alphabet can correspond to a number, and every number can be represented in the binary system with a punching pattern.

Generalisation: Recognising a doubling pattern of 1, 2, 4, and so on, and adding different powers thus generated gives us the binary representation, and we conclude that any number can be represented in terms of powers of 2.

Activity Description: Punched Cards

Consider a set of cards with the alphabet written on them, and we shuffle them. The objective now is to arrange them in alphabetical order. One method is to find the card A first, then B, and so on. This might take some time.
Now, notice that the cards have been punched on the top. There are an O-shaped and a U-shaped punch, and using these holes, we can sort the cards in an interesting way. Insert a stick in the set of cards starting from the rightmost hole. Once you lift the stick, some will come along with it. Now, bring them in front & repeat this for all sets of holes. Magically, the cards are sorted.

In our day-to-day life, we use the decimal number system, which uses the numbers 0, 1, 2, 3, 4, 5, 6, 7, 8, and 9. The decimal number system is based on bundles of ten (Powers of 10, which is 1,10,100,1000, 1000,...). Similarly, the binary number system is based on bundles of two (Power of 2, which is 1,2,4,8,...). So if we compare the two numbering systems for place values, this is what it looks like….

	Decimal (Bundles of Ten)	Binary (Bundles of Two)
Units Place	1	1
First Bundle	10 = 10	2 = 2
Second Bundle	10x10 = 100	2x2 = 4

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Third Bundle	10x10x10 = 1000	2x2x2 = 8

The binary number system (popularly known as the language of computers) deals with only two numbers: 0 and 1. We can convert any decimal number to its binary equivalent and vice versa. You can convert any decimal number into a binary representation by writing it with the power of 2.
for example, 23 = 16 + 4 + 2 + 1. Show this to the students.

23 in decimal	$10^1 = 10$	$10^0 = 1$
23	2	3

23 in binary	$2^4 = 16$	$2^3 = 8$	$2^2 = 4$	$2^1 = 2$	$2^0 = 1$
$(10111)_2$	1	0	1	1	1

You can also do this by an algorithm, which is presented below (which you may present to the student):

i. Divide the decimal number by the base (2) of the binary number.
ii. Note the remainder from the first step till the end of the division process.
iii. The remainders obtained as 0 or 1 are written in order from the last step to the first step of division.

Let's find out the binary equivalent of the decimal number 11.

Divisor	Decimal Number	Corresponding Remainder
2	11	1
2	5	1
2	2	0
	1	

The highlighted cells represent the binary representation, written from bottom to top. So, 11 = $(1011)_2$. The binary number 1011 can be changed to its corresponding decimal number as:

A	B	C	D
2^3 x 1	2^2 x 0	2^1 x 1	2^0 x 1

Decimal number = A+B+C+D = 1 x 2^3 + 0 x 2^2 + 1 x 2^1 + 1 x 2^0
= (1x8)+ (0 x 4) + (1 x 2) + (1 x 1) =8+2+1 = 11

Q.What will be the binary representation for the decimal number 14?

a) 1010
b) 1100
c) 1110

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d) 1111

Explanation

c) 1110
Decimal 14 can be written in power of 2s as 8 + 4 + 2 = 1 x 2^3 + 1 x 2^2 + 1 x 2^1 + 0 x 2^0 = 1110

CT Competency
Algorithmic Thinking
Decomposition
Generalisation

The number of digits or bits (**bi**nary digi**ts**) in a binary number depends on the number. For example, numbers up to 7 can be represented in a 3-bit binary number, numbers from 8 to 15 in a 4-bit binary number, numbers from 16 to 31 similarly need an additional bit, hence a 5-bit binary number, and so on. There are 26 letters in English; hence, 5 sets of holes on each card for sorting.

Q. You want to perform a similar punched card activity with Hindi alphabets. How many holes are needed on each card?

a) 5
b) 6
c) 8
d) 10

Explanation

b) 6
There are 52 Hindi alphabets. Using 6 bits/holes, we can represent numbers from 0 up to 63.

CT Competency
Abstraction
Logic

Punching Pattern on the Card

Every alphabet can correspond to a number, and every number can be represented in the binary system. The following table contains the corresponding conversion from decimal to binary.

Letter	Corresponding Decimal Number	Corresponding Binary Number
A	1	00001
B	2	00010
C	3	00011
D	4	00100
E	5	00101
..	..	..
..	..	..
Y	25	11001

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Z	26	11010

These binary numbers indicate to us a unique way to punch. Where there is 0, punch an O-shaped hole, and where there is 1, punch a U-shaped hole. For example: 01010 is the binary equivalent of J, so the holes on card J from left to right will be O U O U O.

Q. Can you identify which card the binary pattern 10101 belongs to?

a) P
b) S
c) T
d) U

Explanation

d) U

The alphabet U comes 21st in the order starting from A as 1. Therefore, it is represented by the decimal number 21, and that can be written in binary as 10101.

CT Competency

Abstraction
Logic

Q. The alphabet R comes 18th in the order starting from A as 1. What will the punching pattern be on Card R?

a) UOUOU
b) UUOUO
c) UOOUO
d) OUUOO

Explanation

c) UOOUO

The alphabet R comes 18th in the order starting from A as 1. Therefore, it is represented by the decimal number 18, and that can be written in binary as 10010.

CT Competency

Abstraction
Logic

Punched cards, utilising binary choices, is analogous to their role in powering early computing machines such as IBM sorters and Turing's Bombe. This binary concept has its roots in the study of poetry in ancient India, which was well studied by Acharya Pingala and documented in his work Chandaḥśāstra (~2 BCE).

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